

many parts of the world, was eradicated by 1980 due to a vaccination program carried out by the World Health Organization (WHO). The very narrow host range of the smallpox virus—it infects only humans—was a critical factor in the success of this program. Similar worldwide vaccination campaigns are under way to eradicate polio, the incidence of which has dropped by 99%, and measles. Although an effective vaccine exists for measles, a large outbreak occurred in the Pacific Northwest in 2019, correlated with lower vaccination rates in that region. Effective vaccines are also available to protect against rubella, mumps, hepatitis B, and a number of other viral diseases.

Although vaccines can prevent some viral illnesses, medical care can do little, at present, to cure most viral infections once they occur. The antibiotics that help us recover from bacterial infections are powerless against viruses. Antibiotics kill bacteria by inhibiting enzymes specific to bacteria but have little or no effect on eukaryotic or virally encoded enzymes. However, the few enzymes that are encoded only by viruses have provided targets for other drugs. Most antiviral drugs resemble nucleosides and thus interfere with viral nucleic acid synthesis. One such drug is acyclovir, which impedes herpesvirus replication by inhibiting the viral polymerase that synthesizes viral DNA but not the eukaryotic one. Similarly, azidothymidine (AZT) curbs HIV replication by interfering with the synthesis of DNA by reverse transcriptase. In the past 30 years, much effort has gone into developing drugs to treat HIV. Currently, multidrug treatments, sometimes called “cocktails,” are considered to be most effective. Such treatments commonly include a combination of two nucleoside mimics and a protease inhibitor, which interferes with an enzyme required for assembly of the viruses. Multidrug treatments originally involved taking up to 20 pills multiple times per day but now usually consist of a single daily tablet. Another effective treatment involves a drug called maraviroc, which blocks a protein on the surface of human immune cells that helps bind the HIV virus (see Figure 7.8). This drug has also been used successfully to prevent infection in individuals who either have been exposed to, or are at risk of exposure to, HIV.

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in 1999 and has spread to 49 U.S. states, by 2019 resulting in over 50,000 cases and about 2,000 deaths.

The deadly Ebola virus (**Figure 19.10a**), recognized initially in 1976 in central Africa, is one of several emerging viruses that cause *hemorrhagic fever*, an often fatal illness characterized by fever, vomiting, massive bleeding, and circulatory system collapse. In 2014, a widespread outbreak, or **epidemic**, of Ebola virus occurred. By 2016, it had resulted in over 11,000 deaths. In 2017, 2018, and 2019, smaller outbreaks occurred in the Democratic Republic of the Congo.

The mosquito-borne chikungunya virus (**Figure 19.10b**) causes an acute illness with fever, rashes, and persistent joint pain. Chikungunya has long been considered a tropical virus, but it has now appeared in Italy, France, and Spain. The Zika virus (**Figure 19.10c**) was first observed in Uganda in 1947, but for decades only a few cases occurred per year. In the spring of 2015, however, it became an emerging virus when it caused a large outbreak in Brazil. Although symptoms of Zika are often mild, the infection of pregnant women was correlated with a striking increase in the number of babies born with abnormally small brains, a condition called microcephaly. Zika is a mosquito-borne flavivirus (like West Nile virus) that infects neural cells, posing a particular danger to fetal brain development.

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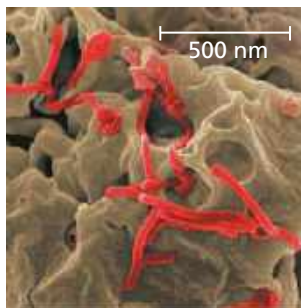


Where do these new strains of virus come from? One cause of rapidly emerging viral diseases in humans is mutation of existing viruses into new viruses that can spread more easily. RNA viruses have a high rate of mutation because viral RNA polymerases do not proofread and correct errors in replicating their RNA genomes. Some mutations change existing viruses into new viral strains that can cause disease, even in people immune to the original virus. A well-known related example is how three or four mutations causing changes in a surface protein of a cat virus (feline panleukopenia virus) resulted in the emergence in 1978 of canine parvovirus, a very contagious deadly virus infecting dogs.

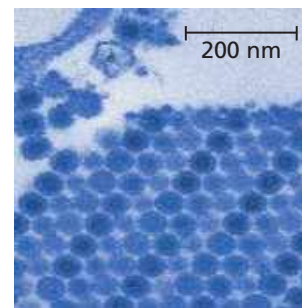
Emerging Viral Diseases

Viruses that suddenly become apparent are often referred to as *emerging viruses*. HIV, the AIDS virus, is a classic example: It appeared in San Francisco in the early 1980s, seemingly out of nowhere, although later studies uncovered a case in the Belgian Congo in 1959. Some other dangerous emerging viruses cause encephalitis, inflammation of the brain. One example is the West Nile virus, which appeared in North America

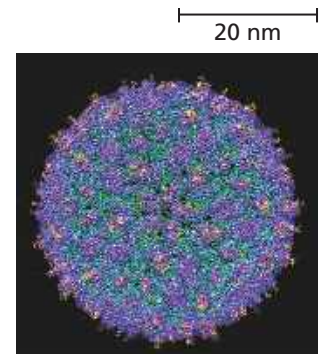
▼ **Figure 19.10** Emerging viruses.



(a) **Ebola viruses** budding from a monkey cell (colorized SEM).



(b) **Chikungunya viruses** emerging from a cell in the upper left and packing together (colorized TEM).



(c) Computer-generated image of a **Zika virus**, based on a technique called cryo-electron microscopy.